Analysis of Energy Inefficiency in Chicago

James Harren (GitHub user: james-harren) - Section 1

2026-03-05

Background

The City of Chicago values energy efficiency as a means to decrease carbon emissions and improve comfort of homes for Chicagoans. The city's Climate Action Plan asserts that carbon emissions from existing buildings are substantial at 69% of total emissions, and energy efficiency programs are important to prioritize..

Already the city, through the Building Energy Use Benchmarking Ordinance, requries buildings over 50,000 square feet to report energy use as an energy efficiency initiative.

The city's Department of Housing adminsters a grant program that aims to help retrofit chicago homes, the Green Homes Chicago Program which provides low income homeowners free energy efficiency upgrades. Future iterations of the program may benefit from different program designs to achieve more energy efficiency benefits per grant dollar. Additionaly, it is important to know if the target grantee is the most 'efficient' target in terms of energy efficiency benefits that can be achieved per dollar.

Research Question

What kinds of neighborhoods have the worst energy efficiency and what areas have the greatest energy efficiency outreach potential?

Approach

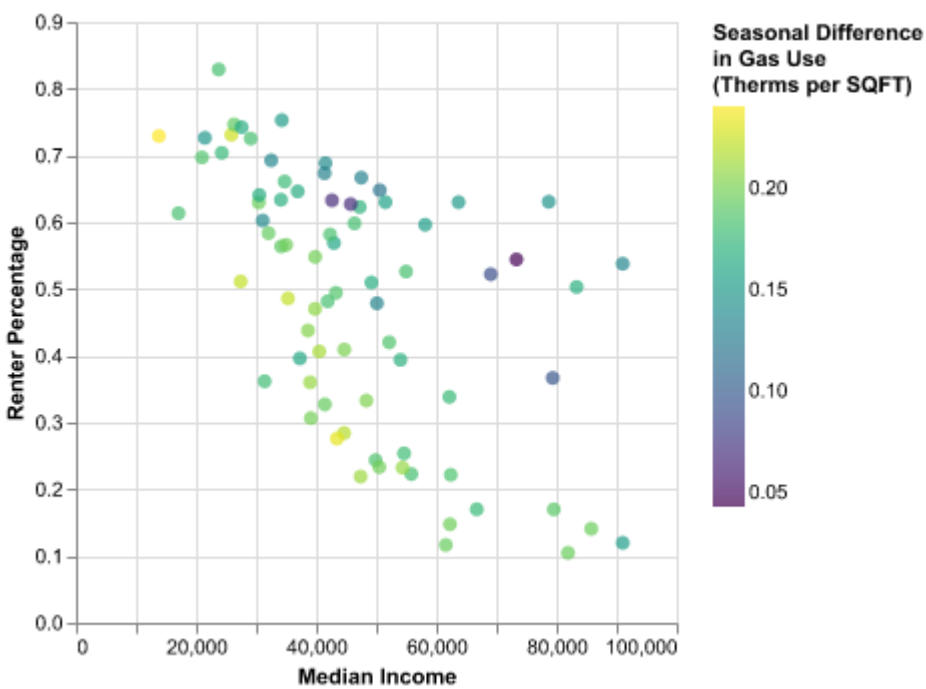
In order to understand what kinds of groups may benefit from energy efficiency programs the most, energy data from the City of Chicago and income data from the American Community Survey from the U.S. Census were analyzed at the Community Area level. Chicago has published energy use data from 2010 aggregated to the census block level and includes both electricity and gas use aggregated to the census block level and information on the buildings within the census block.

Different energy efficiency metrics were calculated at the census block level. First, in order to estimate how insulated a building is, a difference in seasonal gas usage metric was calculated by taking the difference between January, the highest average gas use month, and August, the lowest average gas use month, and dividing by the square footage of the building stock in the census block to be able to compare across geographies. Second, in order to measure how inefficient appliances are, total electricity use for the year is divided by the square footage of the building stock int he census block to compare across geographies.

Data were then combined with income data and aggregated to the community area level. The mean of energy efficiency metrics and average building age were weighted on square footage, and percent renter population and income were weighed on population. Finally, geography of the community areas were merged into the final dataset.

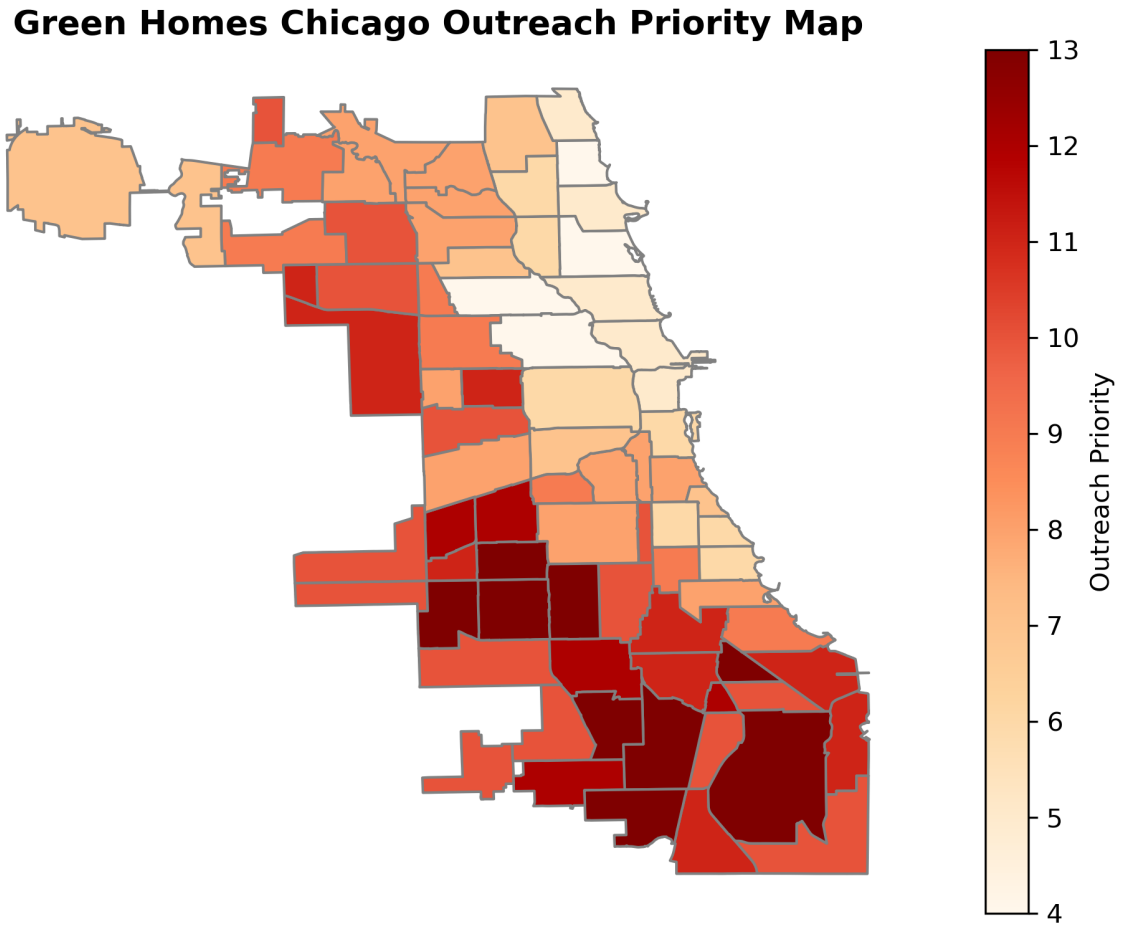
Outcome

So, what neighborhoods does the Green Homes Chicago Program target, and do these neighborhoods have the worst energy efficiency? The following scatter plot shows median income against the percent of renter population for each community area. The plot shows that as renter population deacreases income tends to increase. Additionally, the color shows a clear pattern that worse energy efficiency outcomes, as indicated by the seasonal difference in gas use, occur more often for lower income communitites, but there does not seem to be a correlation on renter composition of community areas.



A simple scatter plot of renter composition and the energy efficiency metric, provided in the dashboard below, shows that there is in fact a correlation between homeownership and poor energy efficiency. It appears that the Green Homes Chicago program appropriately targets grantees!

The neighborhoods that would benefit the most from these programs are on the south and west sides based on a priority score. More information on how the priority score is calculated can be found in the dashboard below.



Streamlit App

A dynamic dashboard allows officials to explore different constructions of energy efficiency plans.

Check out the dashboard here!